



Technical Architecture & Content Pipeline: The "Student Optimum" Blueprint

Context: This project is built by a solo student founder. Time and budget are strictly limited. There is no time to build or maintain a custom Admin Dashboard/CMS. The entire content pipeline must be zero-friction, relying on AI batch generation and direct database injection.

1. The Database Architecture (The "Single-Table" Strategy)

The Solution: A NoSQL approach inside PostgreSQL (Supabase). We will use exactly **ONE** table, leveraging a "Data Payload" column.

Conceptual Schema:

- **Unique ID:** Auto-generated.
- **Target Exam Array:** e.g., [NORCET, ESIC, RRB] .
- **NORCET Stage Tag:** Prelims , Mains , or Both .
- **Type Flag:** standard_mcq , image_mcq , dosage_math , flashcard .
- **Difficulty/Level:** Entry , Mid , High-AIIMS (Crucial for filtering).
- **Category Tag:** Cardiovascular , Community_Health , etc.
- **The Data Payload:** A flexible container holding question, options, answer key, detailed rationale, and media URLs.

2. The NORCET "Prelims vs. Mains" Content Strategy

- **Prelims (Speed & Breadth):** 80 Nursing + 20 General Aptitude. Focus on rapid-fire recall and factual application.
- **Mains (Depth & Judgment):** 100% Nursing. Deep case studies, priority frameworks (ADPIE/ABCs), and multi-step clinical judgments.

3. The Pan-India Expansion Strategy

- **ESIC & RRB:** Focus on general sciences (Biology/Chemistry) and direct one-liner nursing questions.
- **DSSSB:** Heavy focus on the unique 200-mark pattern (Language, Arithmetic, Reasoning).
- **CHO / NHM:** Focus on Public Health, National Programs, and Immunization.
- **Specialty Institutes (NIMHANS):** Deep-dive psychiatric and neurological content.

4. Strategy for Diverse Content Types

- **Topic-Wise & PYQs:** Generate by system. Use "Source" tags for PYQs to display "★ PYQ" badges.
- **IBQs (Image-Based Questions):** Use the "URL Rule"—store images in Supabase Storage buckets, inject the public URL into the JSON.
- **Dosage & Math:** Rationale MUST break down the step-by-step formula.
- **Factual Topics (Flashcards):** Bypass MCQs for lab values/BMW bins. Use a front/back card UI.

5. The Zero-Friction Ingestion Pipeline

1. **AI Batching:** Claude generates 10-20 questions at a time in structured JSON.
2. **Strict Output Enforcement:** AI outputs ONLY the JSON payload.
3. **Direct Injection:** Paste directly into Supabase Dashboard -> "Insert Row."
4. **Instant Live:** Frontend pulls the payload and renders instantly.

6. The AI Content Generation Protocol (The "High-Level" Mandate)

To ensure questions are **High-Level** and **Exam-Relatable**, the generating AI must follow these strict operational rules:

A. The "High-Level" Standard

- **Clinical Reasoning vs. Rote:** The AI is strictly forbidden from generating simple definitions. Every question must force the user to make a *clinical decision*.
- **Priority Frameworks:** Every rationale MUST explicitly state which priority framework was used (e.g., "The nurse uses the ABC framework (Airway, Breathing, Circulation) to prioritize...")
- **NCLEX-Style Complexity:** For NORCET/Mains level, distractors must be clinically plausible. There should be no "obviously wrong" answers; all options should look like legitimate nursing actions, forcing the student to identify the *most correct* one based on clinical acuity.

B. The "Exam-Relatable" Profiles

When generating, the AI must apply the following "Exam Profile" to the batch:

- **NORCET Profile:** NCLEX-style prioritization, ADPIE focus, multi-step clinical scenario, ICU/ER focus.
- **RRB Profile:** Direct questions, heavy biology/science fundamentals, pharmacology side-effects, less prioritization, more factual recall.
- **ESIC Profile:** Mix of general awareness and direct nursing practice. Slightly more direct than NORCET.
- **DSSSB Profile:** Balance of standard nursing with mandatory Arithmetic and Language reasoning questions.

C. Zero Hallucination Protocol

- If citing a guideline, it must reference a verifiable Indian authority (MoHFW, National Immunization Schedule, BMW 2016 Rules) or a global gold standard (WHO, AHA, CDC). No hypothetical page numbers allowed.

D. Formatting Strictness

- The AI must output *perfect* JSON. If a single comma or bracket is missing, the injection fails.
- **Quality Check:** Before generating a batch, the AI must evaluate: "*Is this question actually solvable by a nurse on a ward?*" If it's too abstract or theoretical, discard and re-generate.